$$\text{Relative atomic mass} = \frac{(\text{\% of isotope 1} \times \text{mass number of isotope 1}) + (\text{\% of isotope 2} \times \text{mass number of isotope 2})}{100}$$

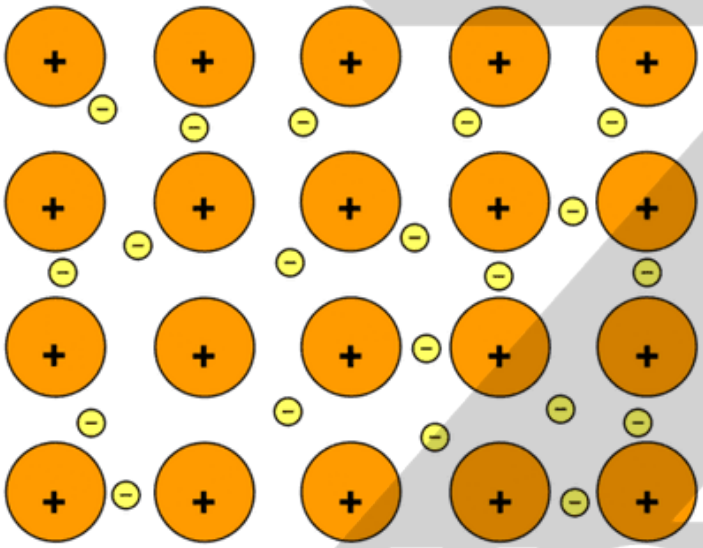
An example for calculating the relative mass and abundance:

- - **Question:** Iridium has two isotopes. These isotopes are Iridium - 191 and Iridium - 193. A natural sample consists of 37.3% of Iridium - 191. Calculate the relative atomic mass (A_r) of the natural sample of Iridium.
    - $\text{\%of Iridium} - 191 = 37.3\%$
    - $\text{\%of Iridium} - 193 = 100\% - 37.3\% = 62.7\%$
    - $A_r = \frac{(37.3 \times 191) + (62.7 \times 193)}{100}$
    - $\therefore A_r$ = 192.2

1.8. Metallic Bonding

Metallic Bonding: the electrostatic attraction between the positive ions in a giant metallic lattice and a "sea" of delocalised electrons.

- M^+ = metal cations
- e^- = delocalised electrons



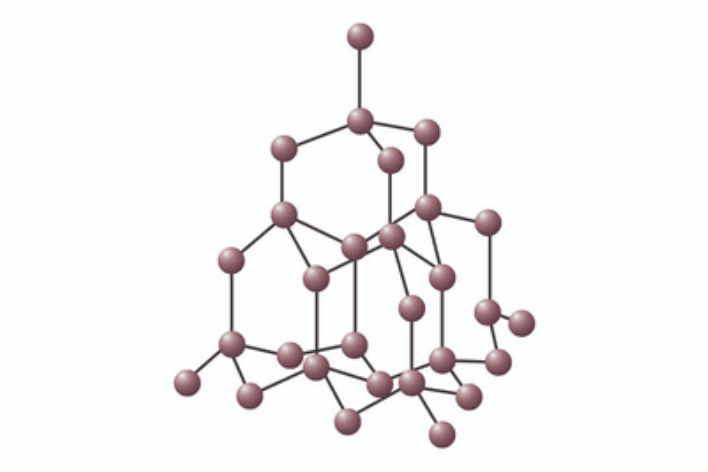
Properties of Metallic Bonding

1. Metallic Bonds have good electrical conductivity: **Delocalised electrons can move through the structures and carry current.**
2. High Melting and Boiling Point: **More energy to overcome strong forces of attraction between positive metal ions and the sea of delocalised electrons and vibrate/transfer heat**
3. Malleability: Can be **hammered into shapes** as **layers can slide over each other.**
4. Ductility: Can be **drawn into thin wires**

1.9. Giant Covalent Structures

Giant Covalent (Macromolecular) Structures: solids with very high melting points, where all the atoms are made of non-metals, mostly including carbon, silicon and oxygen, held together by continuous strong covalent bonds.

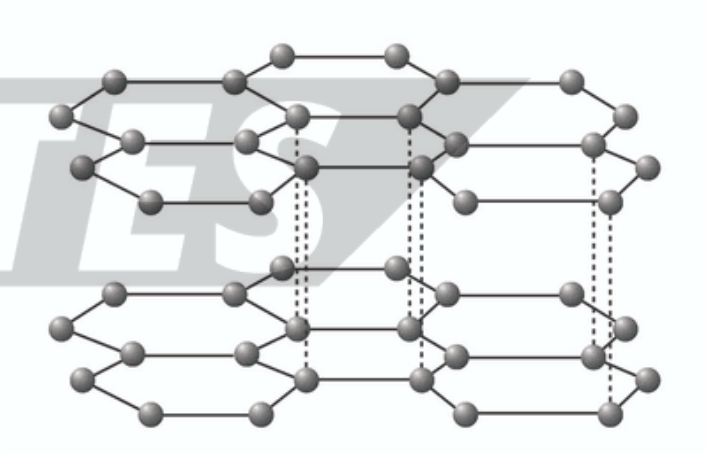
Diamond



Properties

1. Each carbon atom is joined with **four other carbon atoms**
2. **High Melting and Boiling Points** - Strong Covalent Bonds
3. **No Delocalised/Free Moving Electrons**
4. It cannot be scratched easily
5. Transparent colour (Extra information)
6. Cannot conduct electricity **due to no free-moving electrons**
7. **Hard** in structure
8. **Giant Lattice Arrangement**
9. Uses are for **cutting tools**
10. Silicon dioxide has a macromolecular (giant covalent) structure.

Graphite



Properties

1. Each carbon atom is joined with **three other carbon atoms**
2. **High Melting and Boiling Points** - Strong covalent bonds within the layers (but the layers are attracted to each other by weak intermolecular forces)
3. Contains **Delocalised/Free Moving Electrons**
4. It can be scratched easily

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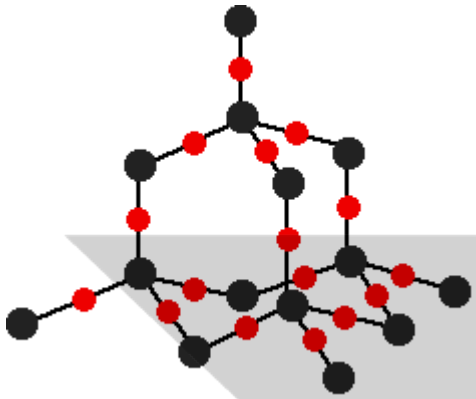
5. Opaque/Black
6. Can conduct electricity **due to free-moving electrons**
7. Soft - **Layers can slide easily**
8. **Layers of hexagonal rings held by weak intermolecular forces**
9. Uses are for **lubricant and electrode** in Electrolysis

Silicon (IV) Oxide



Properties

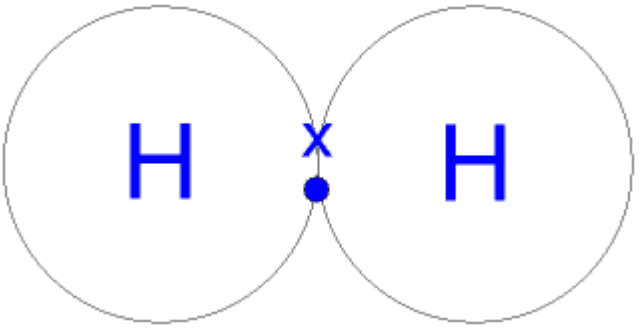
1. The structure of Silicon (IV) Oxide is **similar/resemblance to that of a diamond.**
2. **Hard Structure**
3. **High melting and boiling point** - More energy to overcome
4. **Rigid Tetrahedral Structure**
5. Does **not conduct electricity**

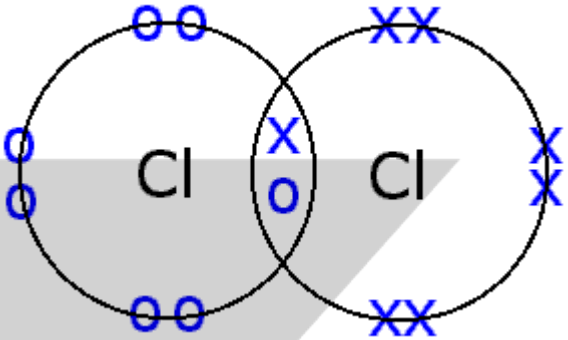
- **Each Silicon atom** is covalently bonded with **4 Oxygen Atoms**
- **Each Oxygen atom** is covalently bonded with **2 Silicon Atoms**

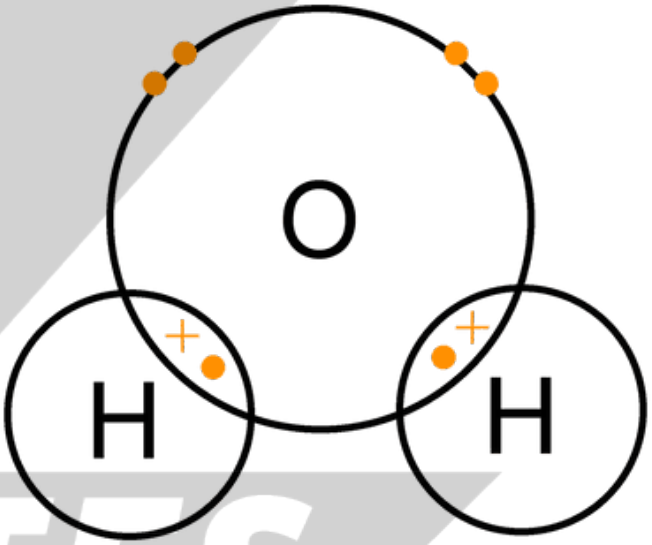
1.12. Simple Molecules and Covalent Bonds

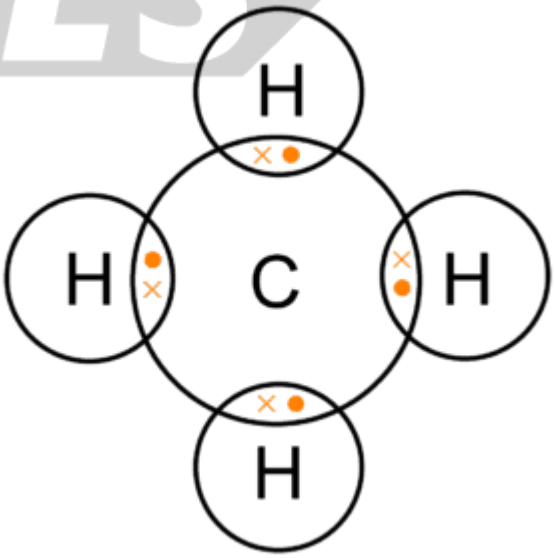
Covalent Bond: Pairs of electrons shared between two atoms leading to noble gas electronic configuration (2.8.8)

Covalent Bonds Dot and Cross Diagram









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